DHRUVA JAAL (■■■■■ ■■■)

Autonomous Zero-Infrastructure Emergency LoRa Mesh, Survivor Client & 3D Tactical Digital Twin

Octo GDG Hackathon · Disaster Management Track
GitHub: **CodeCrafterAdi2006/dhruva_jaal** · License: MIT
ESP32-C3 · 433MHz LoRa · MapLibre WebGL (60FPS) · PWA Web Serial

REAL-WORLD DISASTER GROUND TRUTH (Kathmandu Valley Basin Floods, Sept 27–29, 2024): A 50-year record 322mm rainfall in 24h triggered catastrophic Bagmati/Bishnumati river inundation. Municipal power grid failed; macro cell towers died within 4–6 hours from battery exhaustion, and fiber backhauls were severed. Over 1,000 citizens were stranded on Balkhu/Teku rooftops in pitch darkness with charged smartphones but **0 cellular bars**. Emergency SMS broadcasts failed to deliver; rescuers in rafts navigated blind in fast currents, resulting in 236 fatalities. *Dhruva Jaal asks: What if an off-grid civilian mesh was already there?*

■ 1. Civilian & Dongle Tier

- **$8 Dongle BOM:** ESP32-C3 ($3.20) + Semtech SX1278 ($3.80) + internal helical antenna.
- **USB CDC-ACM Link:** Plugs into Android USB-C port via OTG; draws ~35mA idle listening current.
- **Offline PWA (survivor_app/):** React 18 + Tailwind client running in Chrome via Web Serial / WebUSB API.
- **Zero-Bar UX:** 1-tap SOS, 5s cancel window, medical/trapped flags, and GPS stamping.

■ 2. Mesh & Firmware Tier

- **Flash Store/Forward (store.cpp):** 1.43MB circular flash buffer stores 24,000+ SOS records when nodes drown.
- **Slotted CAD MAC (mac.cpp):** 12.3ms Carrier Activity Detection + jittered backoff avoids collisions.
- **Geographic Forwarding:** Scores neighbors by progress to HQ, link RSSI, and aerial bonus.
- **Two-Way ACK:** Survivor phone turns GREEN only on confirmed receipt from Base Camp.

■ 3. Two-Tier Aerial & 3D Twin

- **Strict 6-Drone Limit:** Airspace deconfliction & battery constraints cap fleet to 6 active units.
- **Steiner Gap Optimizer:** Algorithmic clustering detects isolated survivors & computes bridge points.
- **Auto-Balloon Handover:** If gaps exceed 6, stationed drones are auto-relieved by 100m aerostats.
- **Procedural Audio:** Pure Web Audio API synthesizes real LoRa CSS chirps (800Hz→2400Hz).

RF Physics Parameter	Design Value	Operational Significance
Carrier Frequency	433.00 MHz ISM	λ=69.2cm; superior knife-edge diffraction vs 868/915MHz in wet masonry.
Modulation Profile	SF10 · BW 125kHz · CR 4/5	Receiver sensitivity -137 dBm; link budget +157 dB for urban penetration.
Time-on-Air (ToA)	370.7 ms (32-byte frame)	Sub-400ms airtime minimizes channel occupancy and collision probability.
1st Fresnel Zone (F■)	F■ = 8.656√(d_km / 0.433)	Ground links blocked by flood/roofs (<20%); 100m aerostat guarantees 96% LoS.
Routing Score Metric	0.50·Δd + 0.30·RSSI + 0.20·Alt	Steers packets geographically toward Base Camp without global routing tables.

Timeline	Phase	Simulated Event & System Response
t = 0.0s	P1	Grid blackout; Survivor N1 (Balkhu) sends SOS1 across ground mesh.
t = 12.6s	P1	Base Camp receives SOS1; returns 2-way ACK; N1 turns GREEN.
t = 18.0s	P2	Bagmati floods reach 4.5m; Relay N2 submerged (OFFLINE).
t = 24.0s	P2	N1 sends SOS2; path severed; cached in SPI flash memory (YELLOW).
t = 30.0s	P3	Base Camp scrambles Drone DR1 to 80m; bridges gap & pulls SOS2.
t = 46.0s	P4	Aerostat BL1 ascends to 100m; DR1 freed; Balloon ACK turns N1 GREEN.

Monorepo Structure: **frontend/** (3D Command Digital Twin & Audio Synthesizer) · **survivor_app/** (React 18 PWA + Web Serial API) · **firmware/** (ESP32-C3 Bare-Metal C++ Mesh Driver) · **docs/** (Protocols)

Live Verification: Run cd frontend && npm run dev → **http://localhost:5173** (Space=Play, R=Reset, Mode=Sandbox, Auto=Gap-Bridge). Run cd survivor_app && npm run dev → **http://localhost:5173/?mock** (Phone Simulator). **60 FPS WebGL Guaranteed.**